Amritanshu Kumar

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Professional Summary

Ships full-stack game systems — client, server, API, infrastructure. Primary contributor on a live multiplayer title serving **14K+ players**. Built C++ physics engine and Unity hot-reload tool (**7ms** IL patching) from first principles. **5M+ downloads** shipped.

Professional Experience

RootHoot Pvt Ltd **Remote, Hyderabad**
Team Lead *Oct 2024 – Present*

- Lead Pongz (multiplayer Mahjong, 7-engineer team) serving **2,800+ daily sessions**. Own ~55% of the 400K-line codebase spanning client, server, and API.
- Architected socket framework — enforced connection state machine, typed event dispatch, TTL-expiring offline message queue, parallel-probe connectivity with adaptive polling.
- Designed server-authoritative protocol resolving race conditions from latency and message reordering — monotonic turn sequencing, fingerprint-based deduplication on reconnect, exponential-backoff with jitter.
- Built three-stage AFK system — warning → server-tracked idle → bot substitution mid-match. Full state resync on player return without interrupting remaining three.
- Engineered hierarchical state controller — 8 phases, 14+ substates — replacing boolean flags with typed queries as single source of truth for gameplay, UI, networking.
- Extracted 8 reusable Unity systems (Event Bus, State Machine, Save, Pause, Logging, UI, Loading, Popup) — split event/message bus, WeakReference lifecycle, atomic-write persistence, bit-flag pause — adopted team-wide.
- Led 7 engineers — architecture, code reviews, sprint planning — while remaining primary code contributor.

DeftSoft **Mohali, Punjab**
Game Developer *Feb 2023 – Oct 2024*

- Shipped endless runner — **5M+ downloads**, **4.6 stars** on Google Play. Improved engagement **30%** via player-data-driven iteration.
- Introduced TDD team-wide — ratings improved **1.5 stars**. Authored **25-chapter** technical course on game architecture.
- Built procedural generation tools — **40%** faster level design. Optimized rendering — **25%** higher FPS on low-end devices.
- Mentored 3 junior developers — technical reviews, pair programming, coding standards.

Selected Projects

InstaReload: Unity hot reload — domain reload 3–5s → **7ms**. Two paths: Roslyn debug (90% of edits), MonoMod detours (structural changes).

Nimirta Engine: Custom 2D engine, C++/SFML — impulse solver with warm starting, Baumgarte correction, semi-implicit Euler, fixed timestep, spiral-of-death prevention, 3 collision algorithms, 8 materials, hand-written vector math, 3-tier AI.

FlowUI: Open-source Unity UI framework (MIT) — CRTP builders, Roslyn codegen with partial classes, bidirectional O(1) lookups, 8-tab editor. Full documentation site.

Recurve 28: Hardware-integrated archery — Unity, Arduino, projector, OpenCV. Real-time arrow impact detection.

Technical Skills

- **Languages:** C#, TypeScript, C++, Go, Python
- **Game Engines:** Unity, Unreal Engine, Custom (C++)
- **Backend:** Node.js, Express, MongoDB, Redis, Socket.IO, Docker, Google Cloud Run, Firebase
- **Specializations:** System Architecture, Real-Time Multiplayer, Physics Simulation, State Synchronization, TDD, ECS

Education & Certifications

Lovely Professional University **Punjab, India**
Bachelor of Computer Application *Sep 2020 – Jun 2023*
Certifications: Unity Certified Programmer | Unreal Engine Game Design & Development Specialization